

FOUNDATIONS — WHAT EXPRESSIONS ARE MADE OF

Variable A letter that stands for an unknown number. <i>e.g., x, n, b</i>	Constant A number that does not change. <i>e.g., 7 in $2n + 7$</i>	Coefficient The number that multiplies a variable. <i>e.g., 3 in $3x$</i>
Term A number, variable, or product separated by + or −. <i>e.g., 3, $2x$, $5y$ in $3 + 2x - 5y$</i>	Expression Numbers, variables, and operations combined. <i>e.g., $2n + 5$, $3x - 4$</i>	Evaluate Substitute values and follow the order to find one value.

READING & WRITING EXPRESSIONS

Verbal phrase Words that describe a mathematical situation. <i>e.g., "twice a number"</i>	Key words Words that signal an operation: sum, product, less than, quotient. <i>e.g., sum \u2192 +</i>	Formula A rule written with variables. <i>e.g., $P = 2l + 2w$</i>
Scenario A real-world situation to model with math. <i>e.g., buying apple cider at a stand</i>		

ORDER OF OPERATIONS — HOW TO EVALUATE

Order of operations The agreed sequence for evaluating an expression. <i>e.g., simplify inside first</i>	GEMDAS Grouping, Exponents, Multiply/Divide, Add/Subtract. <i>e.g., parentheses $\rightarrow$ exponents $\rightarrow$ $\times \div \rightarrow + -$</i>	Grouping symbols Parentheses, brackets, and braces. <i>e.g., $5 + (2 \times 3)$</i>
Nested grouping Grouping inside grouping — work innermost first. <i>e.g., $2[(3 + 1) \times 2]$</i>		

SCIENTIFIC NOTATION — POWERS OF TEN

Scientific notation A number written as $a \times 10^n$, where $1 \leq a < 10$. <i>e.g., 3.2×10^4</i>	Power of 10 10^n — ten multiplied by itself n times. <i>e.g., $10^3 = 1,000$</i>	Standard form The usual way of writing a number. <i>e.g., 32,000 vs. 3.2×10^4</i>
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SIMPLIFYING EXPRESSIONS

Like terms Terms with the same variable(s) and exponents. <i>e.g., $3x$ and $5x$ are like</i>	Combine Add or subtract the coefficients of like terms. <i>e.g., $3x + 5x = 8x$</i>	Simplify Write the expression in its shortest form. <i>e.g., $3x + 2x \rightarrow 5x$</i>
Distributive property $a(b + c) = ab + ac$ <i>e.g., $2(x + 3) = 2x + 6$</i>	Expand Remove parentheses by distributing. <i>e.g., $3(x - 4) \rightarrow 3x - 12$</i>	Substitute Replace each variable with its given value. <i>e.g., $x = 4$ in $3x \rightarrow 12$</i>

EQUIVALENCE, AREA & FINANCIAL CONNECTIONS

Equivalent expressions Expressions with the same value for every input. <i>e.g., $2(x + 3) \rightarrow 2x + 6$</i>	Greatest common factor (GCF) The largest shared factor of the terms. <i>e.g., $GCF(6x, 9) = 3$</i>	Factored form The GCF times the remaining terms. <i>e.g., $6x + 9 = 3(2x + 3)$</i>
Perimeter The distance around a shape. <i>e.g., a rectangle, $2l + 2w$</i>	Area The space inside a shape. <i>e.g., a rectangle, lw</i>	Income / Expenses / Budget Income = money in · Expenses = money out · Budget = income minus expenses. <i>e.g., $I - E$</i>